

GTU CSE 241 / 501 Quiz

Std Code: **134**

Std Number:

2	1	0	1	0	4	0	0	4	0	1	7			
---	---	---	---	---	---	---	---	---	---	---	---	--	--	--

```
class Digit {  
public:  
    Digit();  
    Digit(unsigned int);  
    int check-digit(int) const;  
    friend ostream& operator<<(ostream& , const Di  
                                git& );  
    friend Digit& operator++(Digit& ); // prefix  
    friend Digit& operator++(Digit& , int); // postfix  
    friend ostream& operator<<(ostream& os, const Di  
                                git& d);  
    friend const Digit& operator+(const Digit& d1, co  
                                nst Digit& d2);  
private:  
    unsigned int digit;  
};
```

```
Digit ::= Digit
```

```
% digit + (0) {}
```

```
Digit ::= Digit (unsigned int value)
```

```
% Digit + (check-digit + (value)) {}
```

```
int + Digit ::= check-digit + (int value) {}
```

```
if (value < 0) {
```

```
    exit(1);
```

```
}
```

```
    return (value % 10); }
```

```
ofs + reomb operator << (ofs + reomb ofs, const Digit + d) {
```

```
    ofs << d; digit << 1;
```

```
    return ofs;
```

```
}
```

```
Digit + operator ++ (Digit + d) {
```

```
    d + digit + 1;
```

```
    d + digit = check-digit + (d + digit);
```

```
    return d;
```

```
Digit + operator ++ (Digit + d, int ignore) {
```

```
    int temp = d + digit;
```

```
    d + digit ++;
```

```
    d + digit = check-digit + (d + digit);
```

```
    return Digit + (temp); }
```

```
ofs + reomb operator << (ofs + reomb ofs, const Digit + d) {
```

```
    ofs << d; digit << 1;
```

```
    return ofs; }
```

```
const Digit + operator + (const Digit + d1, const Digit + d2) {
```

```
    int temp = check-digit + (d1 + digit + d2 + digit);
```

```
    return Digit + (temp); }
```